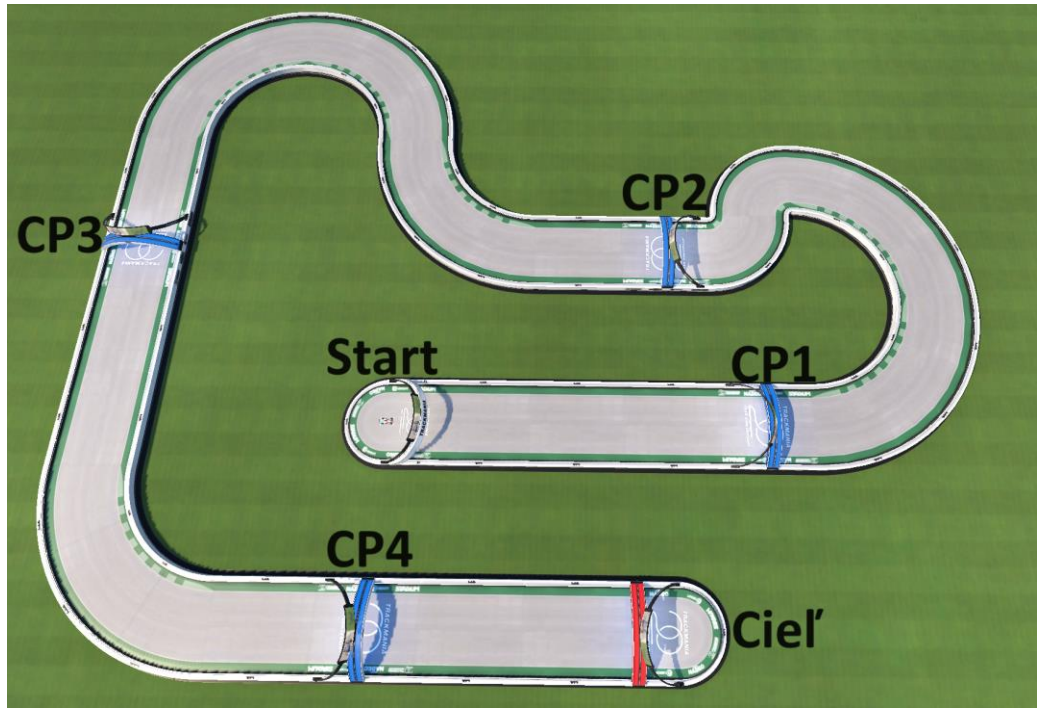
Trénovanie autonómneho vozidla v hre Trackmania

Timotej Melkovič

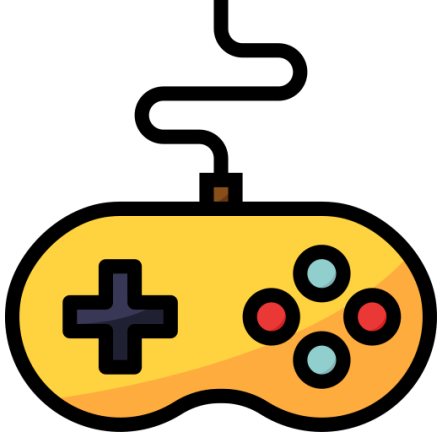
Trackmania



Ciele práce

- Vytvoriť čo nejlepšieho šoféra (agenta)
- Agent – Umelá inteligencia (senzory + akcie)

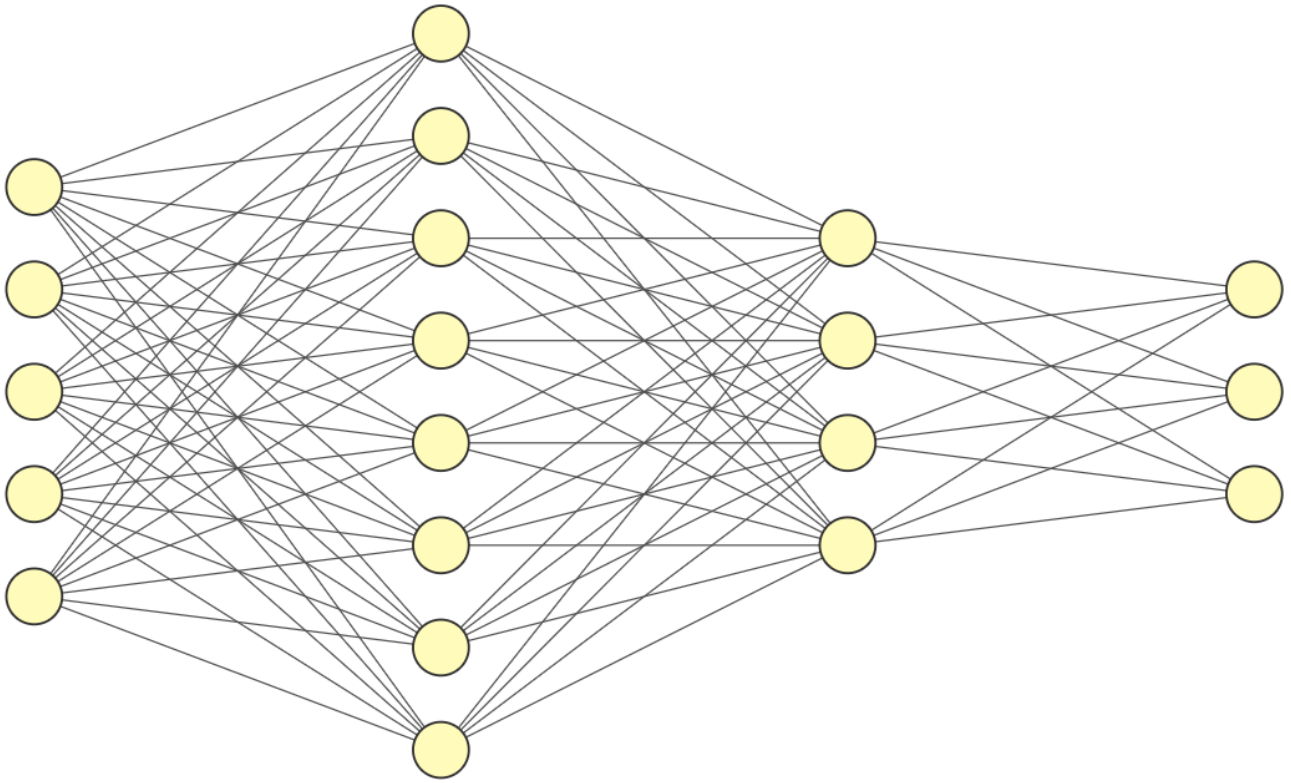
Akcie



- Plyn (0, 1)
- Brzda (0, 1)
- Zatáčanie (-1, 1)

Umelá inteligencia

Hlboká neurónová sieť

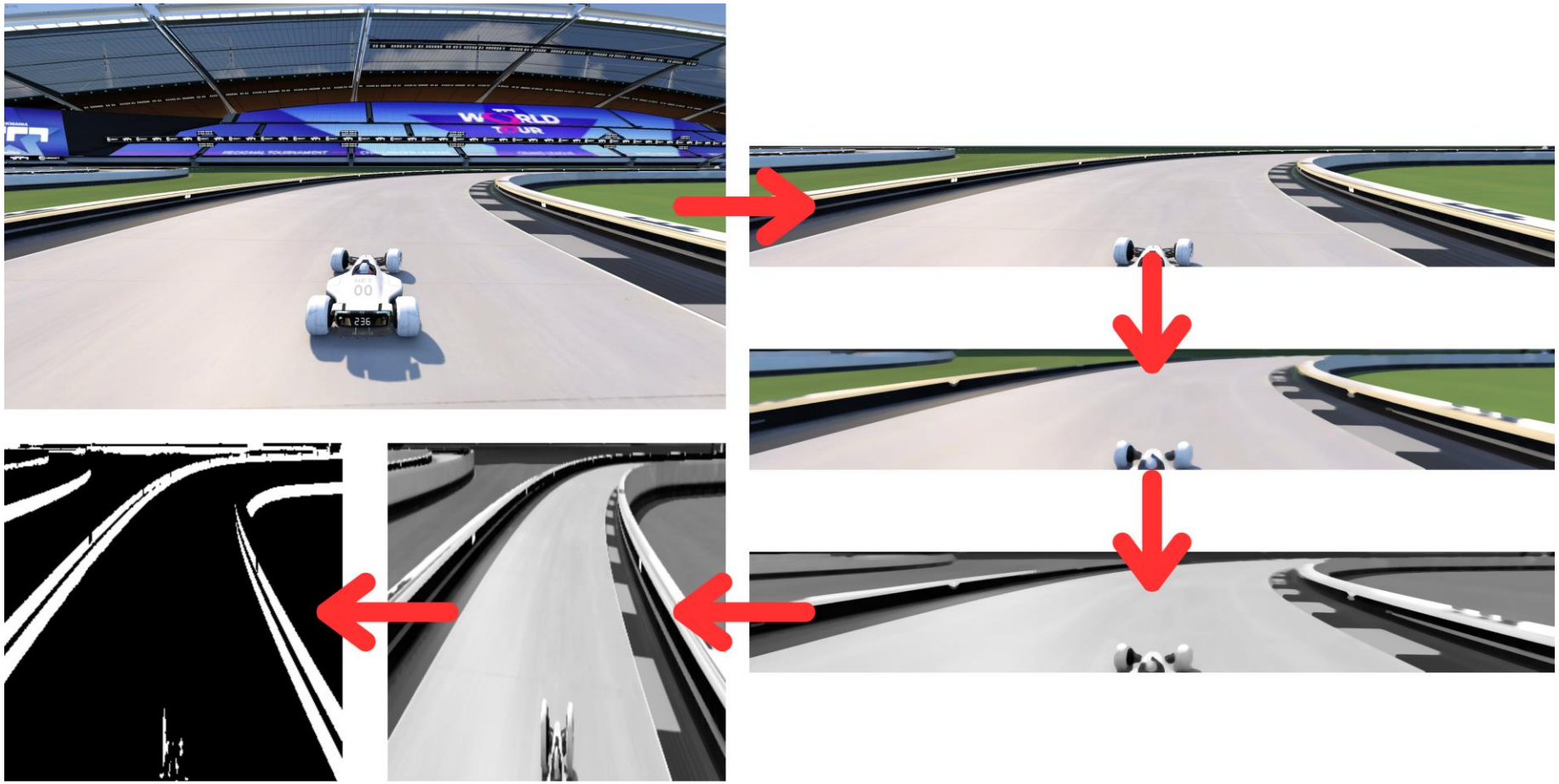


Input Layer $\in \mathbb{R}^5$ Hidden Layer $\in \mathbb{R}^8$ Hidden Layer $\in \mathbb{R}^4$ Output Layer $\in \mathbb{R}^3$

Strojové učenie

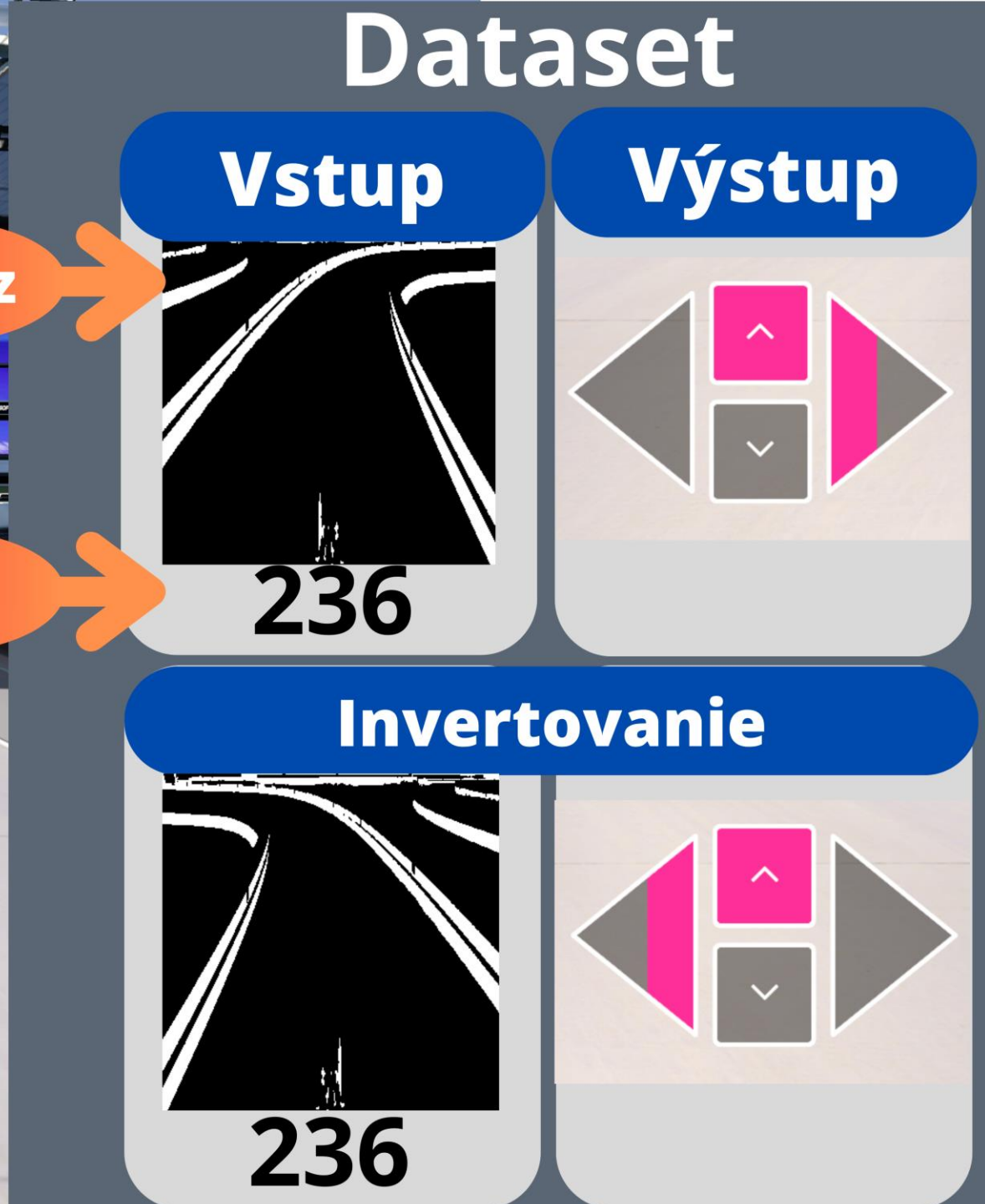
- Učenie pod dohľadom
  - Vstupné dáta s požadovanou výstupnou akciou
- ~~Učenie bez dohľadu~~
- Učenie posilňovaním
  - Agent sa učí ako maximalizovať odmenu

Senzory

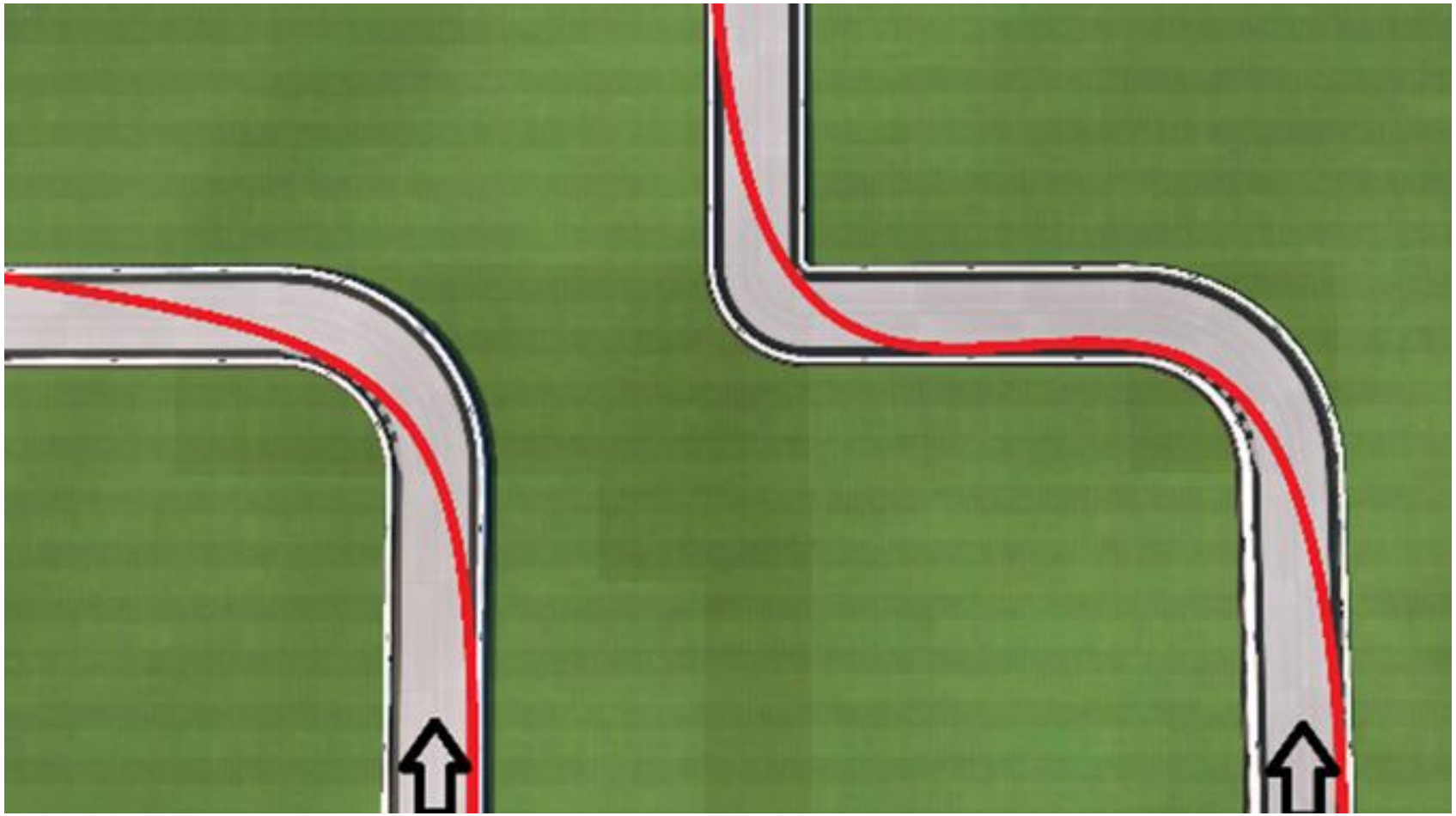


Implementácia

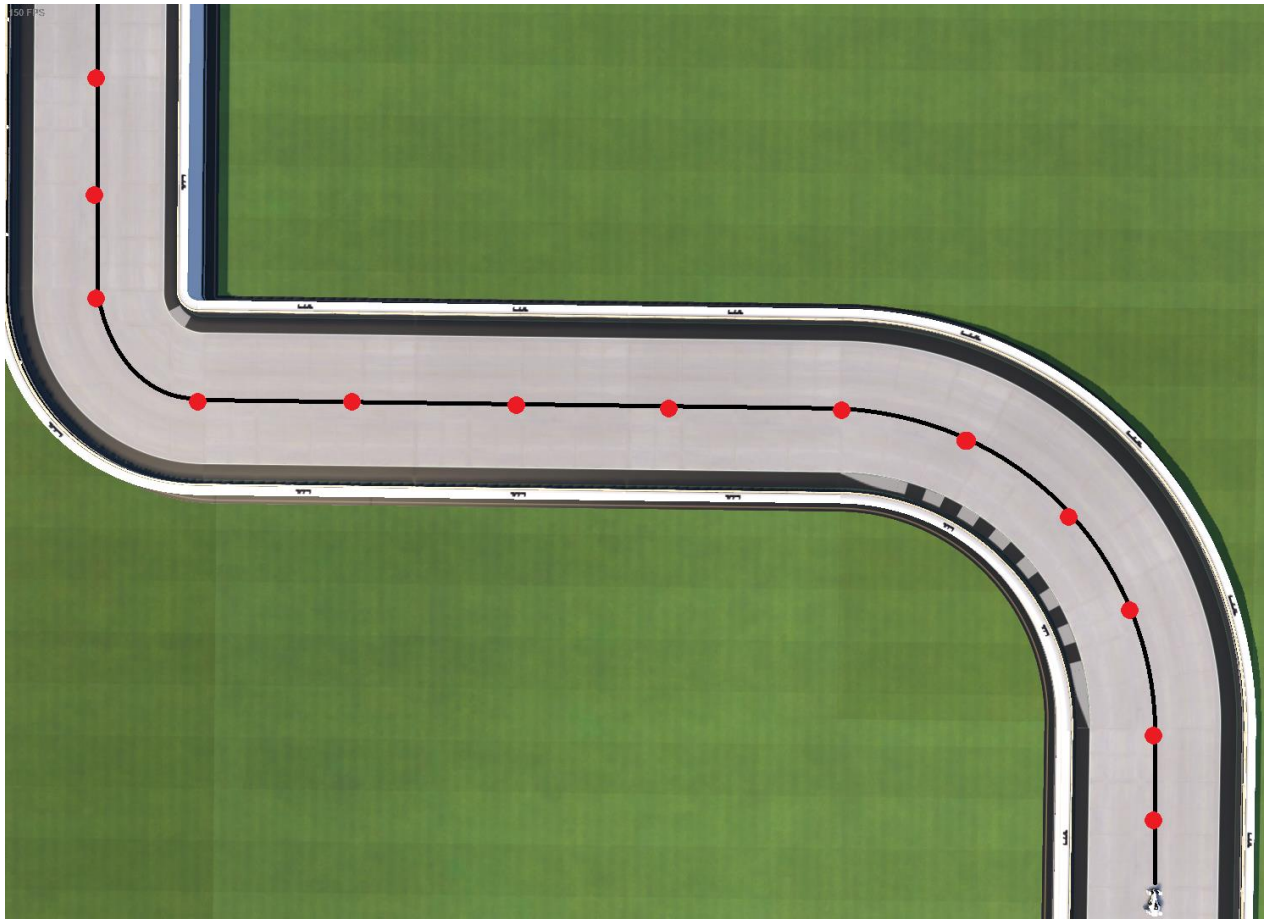
- Učenie pod dohľadom
- Screenshoty z hry



Neznalosť trate

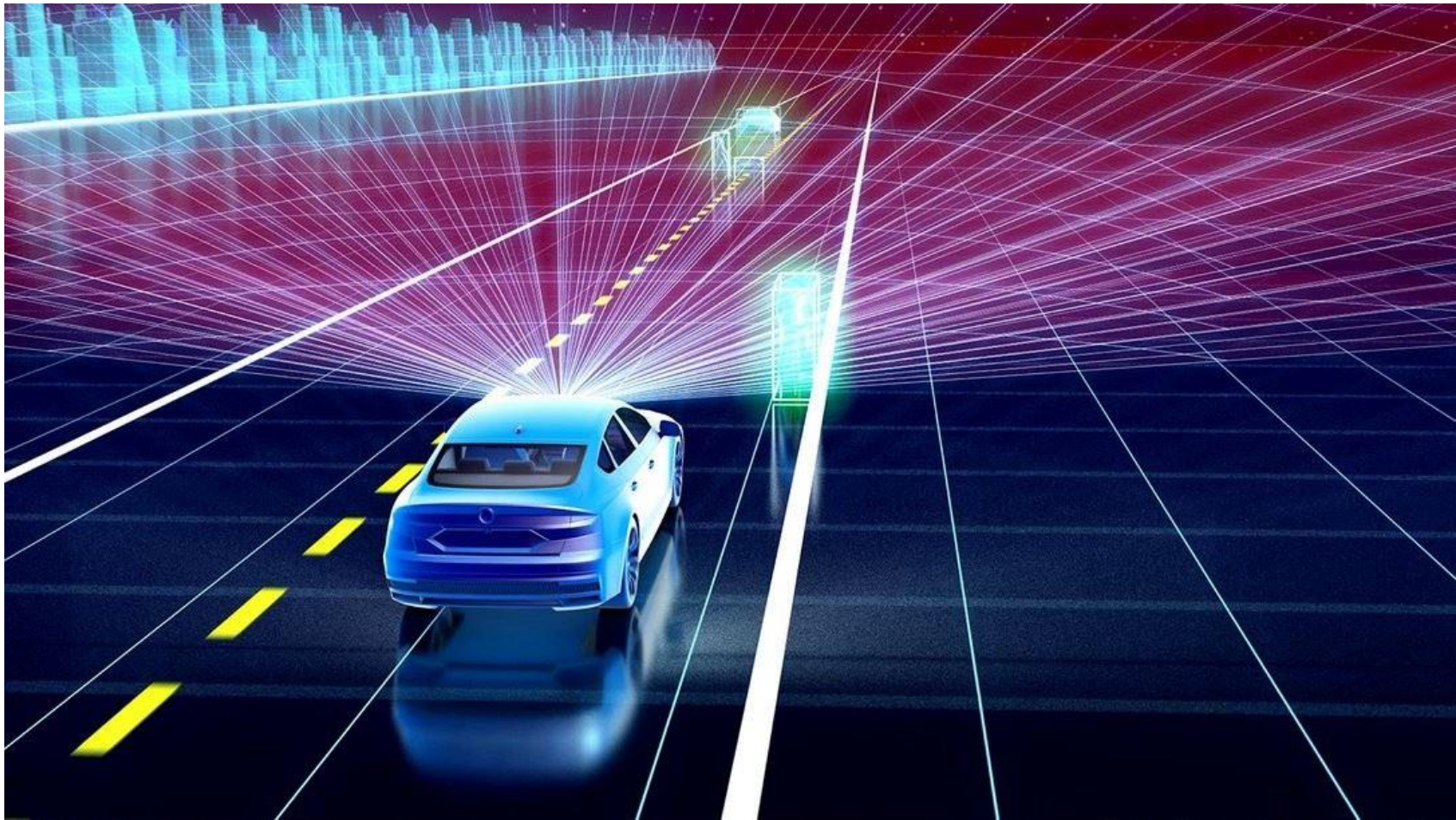


Riešenie

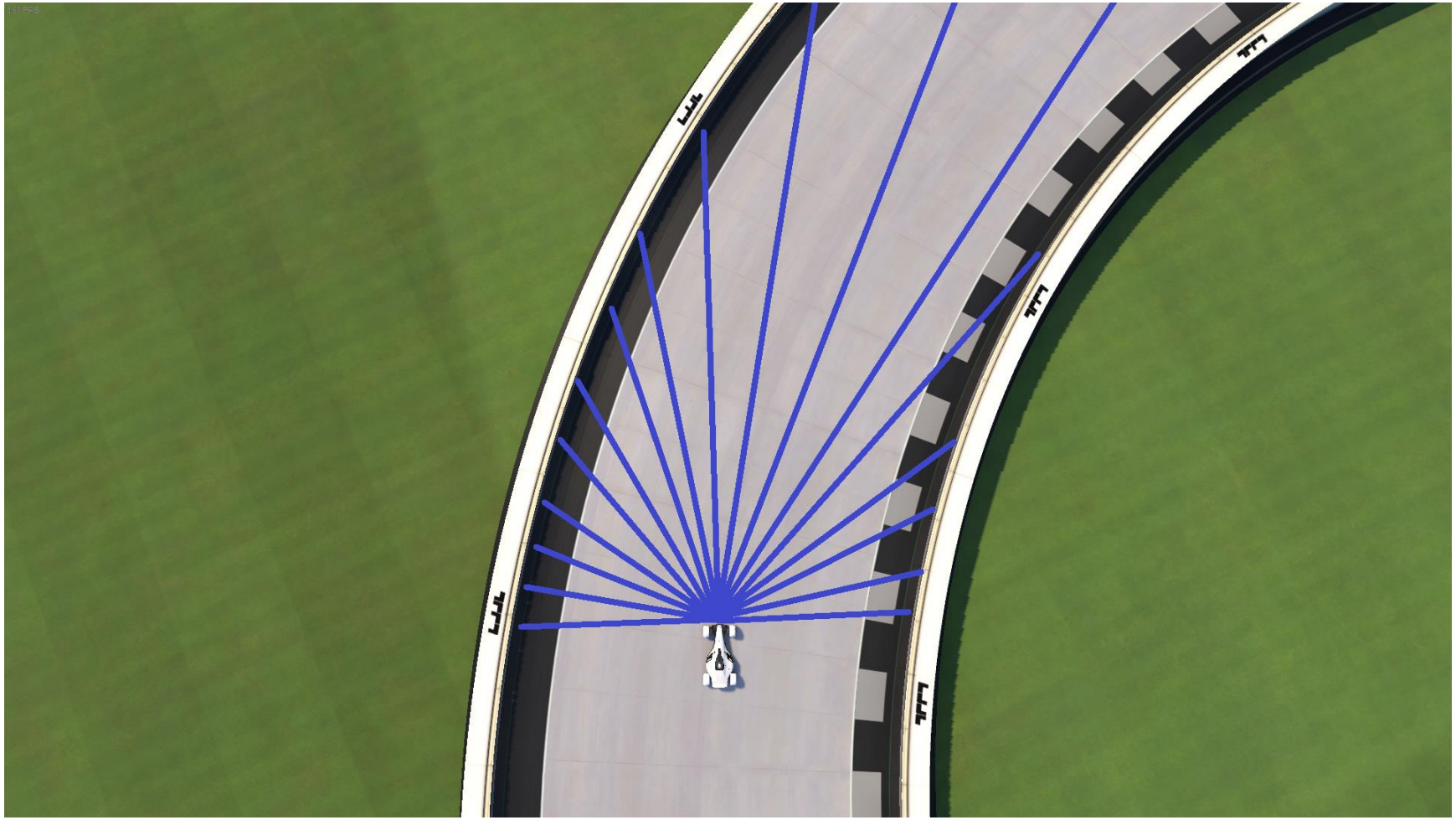


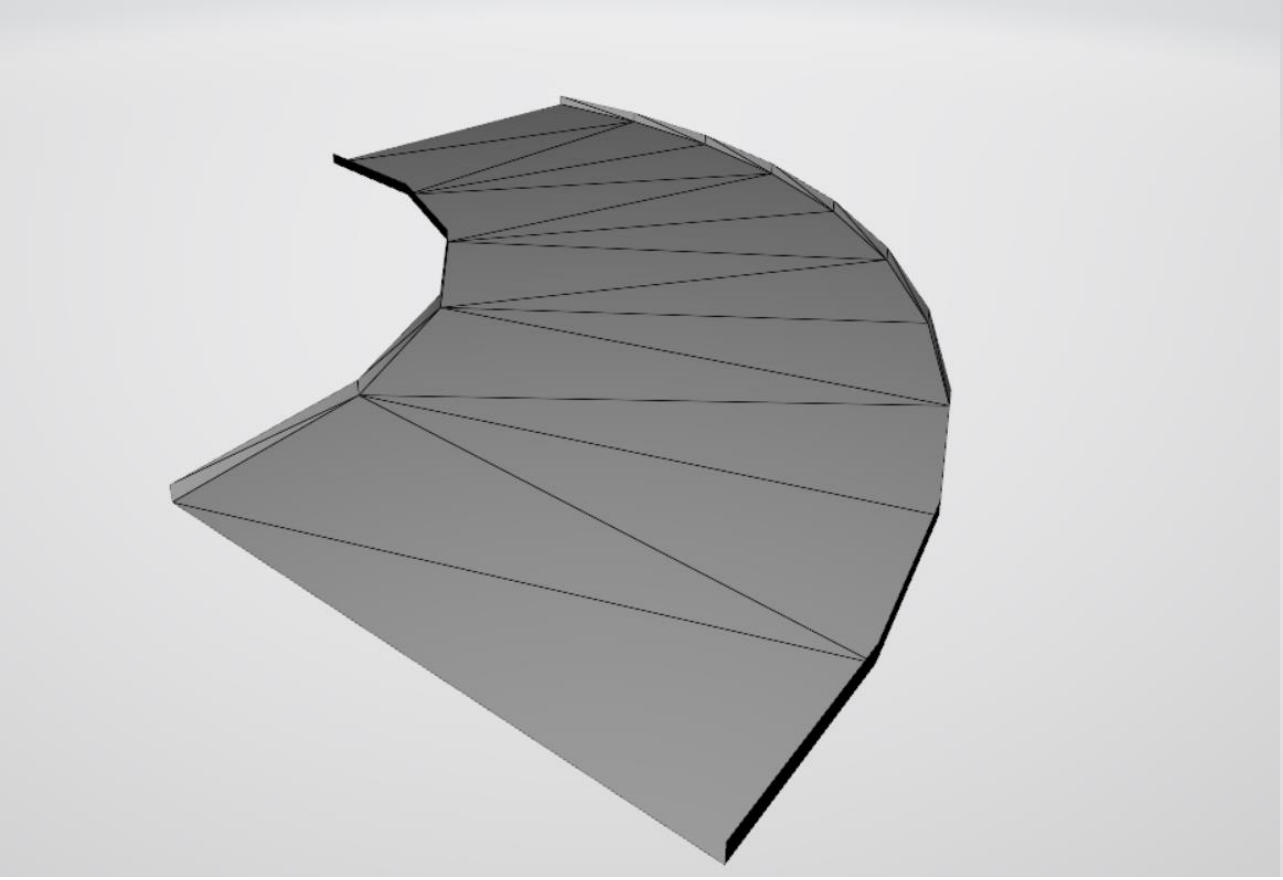
Pomalá odozva screenshotov

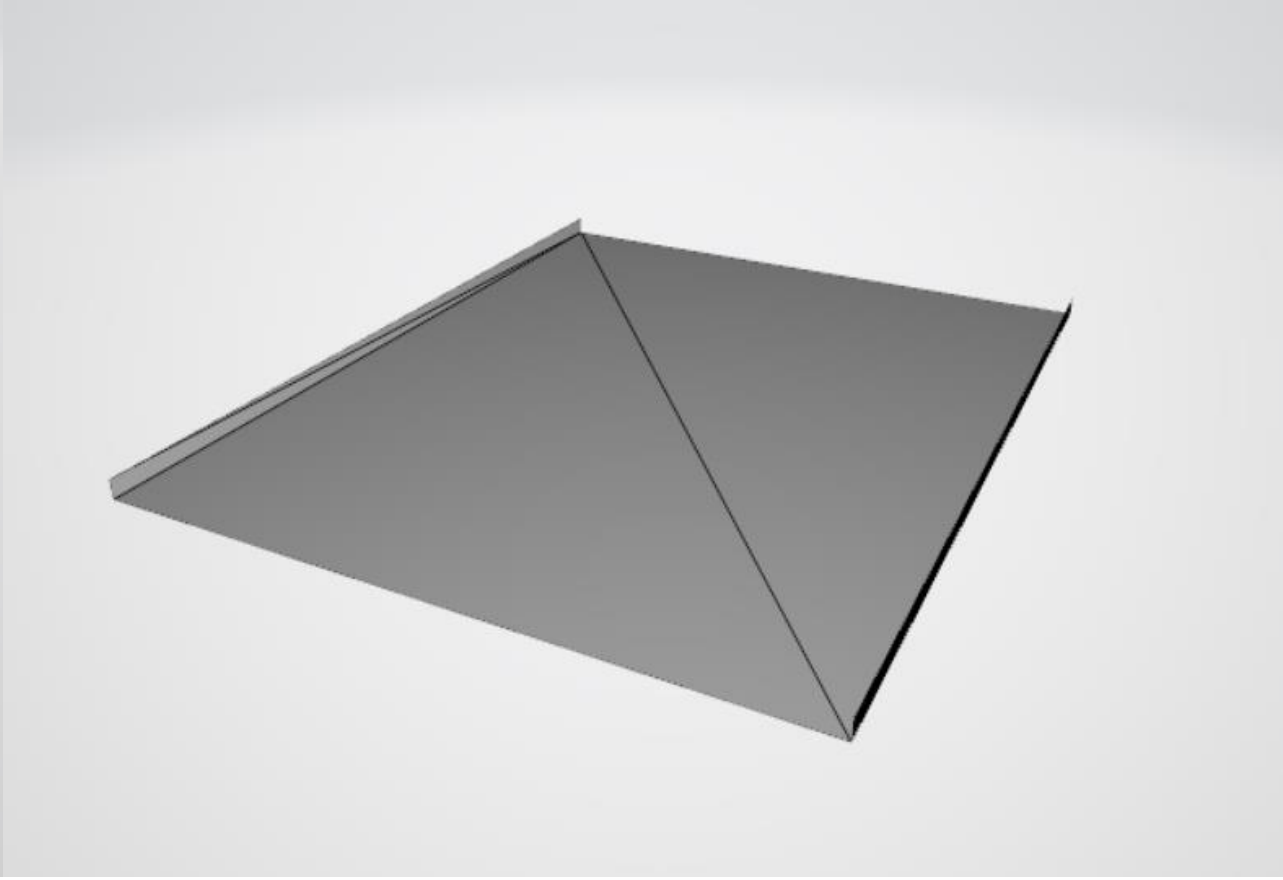
- Veľa informácií
- Šum
- Lidar



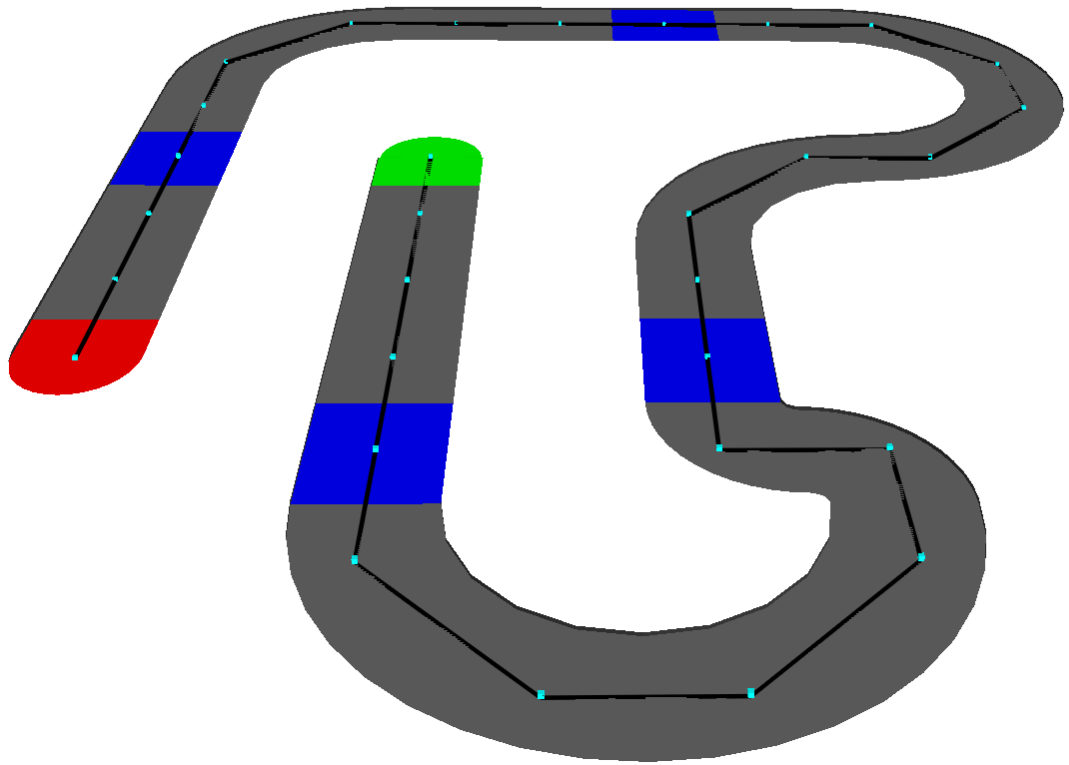
Riešenie



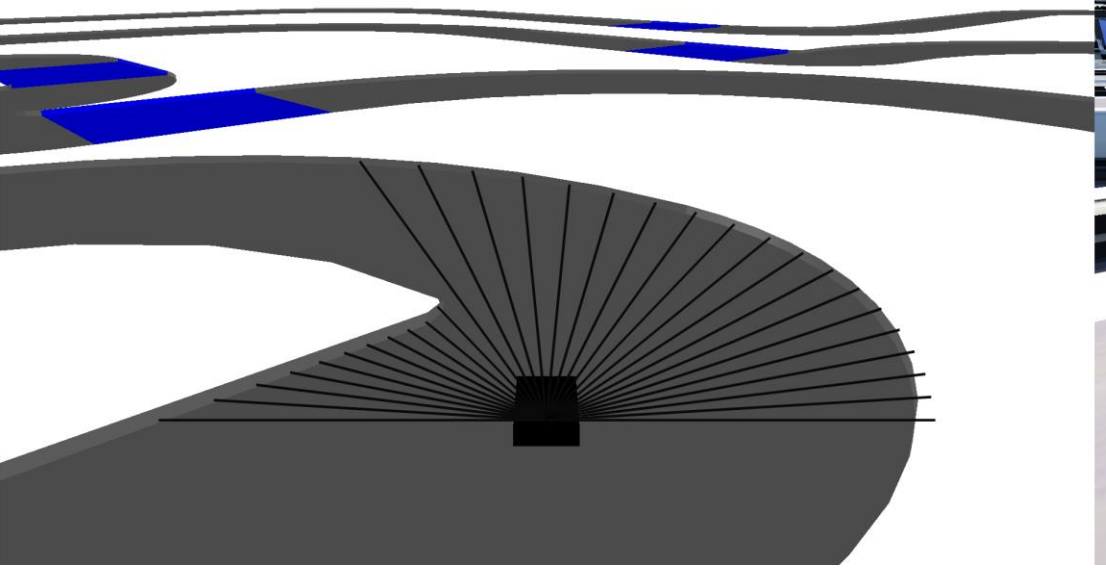




3D Grafika

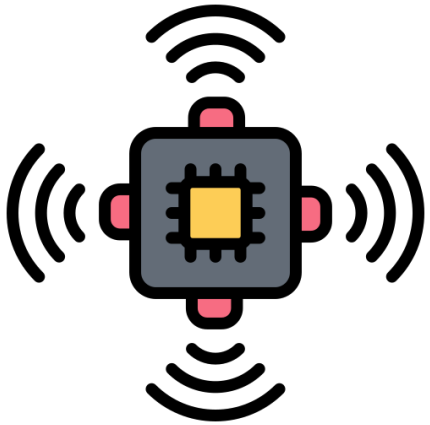


Raycasting

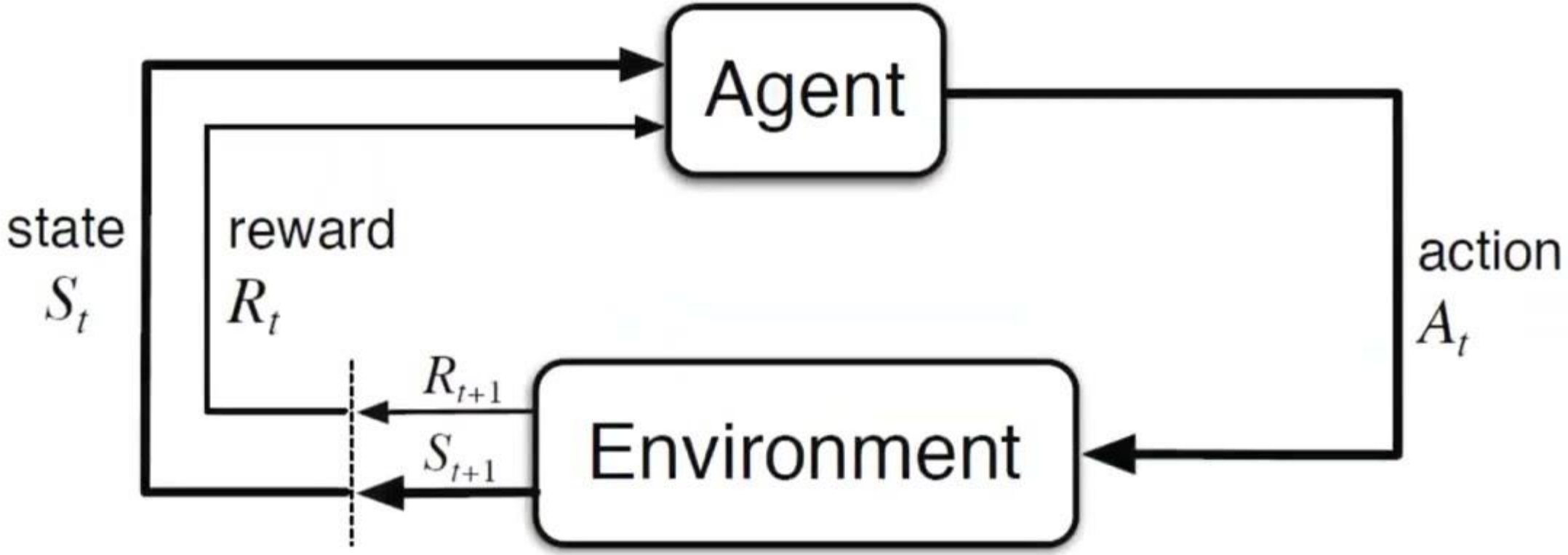


Medal Time
Author 2:41.557
Pers. Best 2:41.678
Gold 2:52.000
Silver 3:14.000
Bronze 4:03.000
MET
01

Senzory znovu



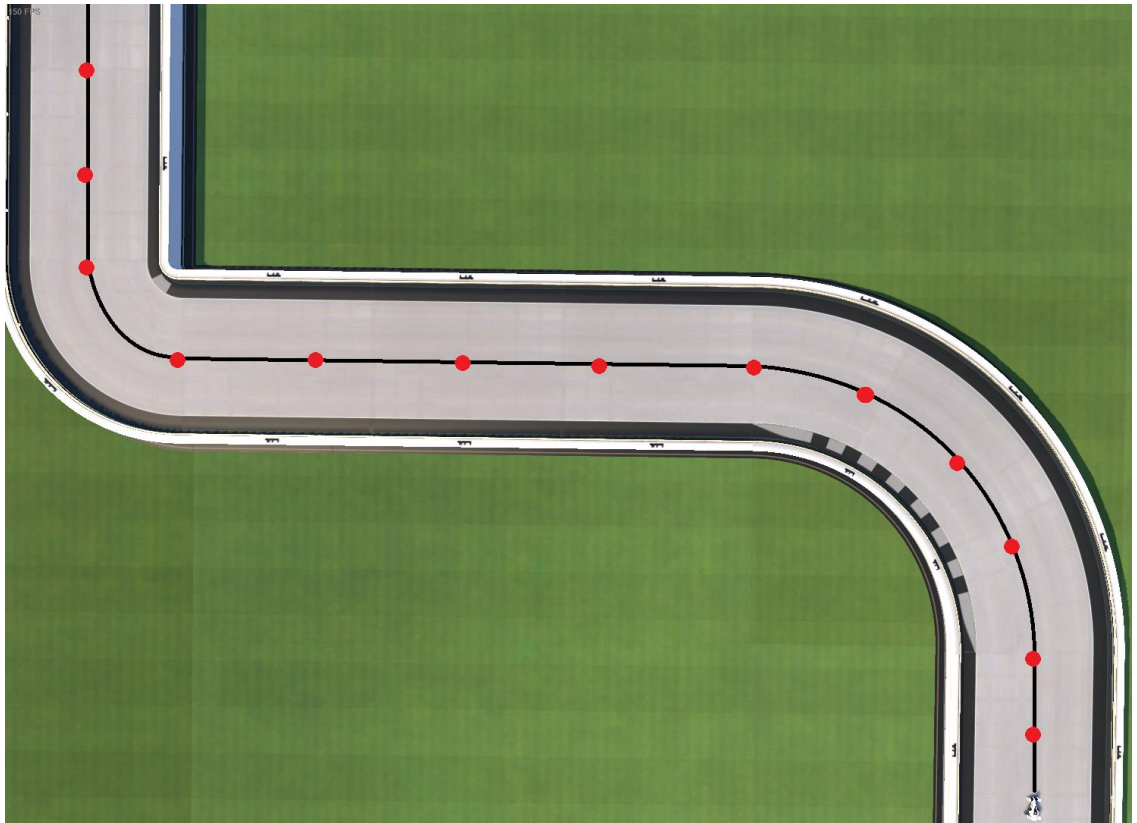
- N vzdialeností z raycastingu
- X, Z pozícia auta
- Dx, Dz otočenie auta
- Rýchlosť auta
- M nasledujúcich blokov trate



Source: *Reinforcement Learning: An Introduction*, Sutton & Barto, 2020

Objective

$$\max_{\pi} \mathbb{E}\left[\sum^{\infty} \gamma^k R_t\right], \quad 0 < \gamma < 1$$

Reinforcment learning

- StableBaselines
- Agent sa snaží maximalizovať reward fuction